Urban Green Space Optimizer: A Deep Learning Approach for Sustainable Plantation Planning Using Image-Based Land Analysis

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***Abstract*—Degradation of green spaces in cities has become a key issue in terms of sustainable development of cities. In urban locations, it can be complicated to find appropriate sites to plant the plantation because there is no automated device that can help analyze the suitability of land. This paper will present a proposal of Urban Green Space Optimizer, an AI-based solution to the problem utilizing deep learning and computer vision to shape potential plantable areas on land image. The proposed system applies the Convolutional Neural Network (CNN)-based segmentation method to identify the plantable regions in the input images. The pixel based spatial estimation is then used to further analyze the segmented regions in order to provide the available plantable area. According to the established ecological spacing limitations, the system determines how many trees can be planted in a particular area in a sustainable way. Web-based interface is built to enable the users to upload pictures and view the identified zones to be planted and the capacity of the plantation. This system was trained on a dataset of 2,279 samples (DeepGlobe and LoveDA) in 50 epochs, and with a best validation IoU of 0.7640. The model got a test IoU of 0.6286, a Dice Score of 0.7254, a Precision of 0.7663, and a Recall of 0.7370 on the held-out test set of 229 images. The framework shows that deep learning and a spatial analysis can be effectively integrated to help plan cities in an environmentally friendly way, and the model trained will be able to provide actionable values of plantable area, which can be used to support decisions in the city as long as it is used in municipal decision support.**

***Index Terms*—Urban green space, deep learning, convolutional neural networks (CNN), image segmentation, sustainable urban planning, computer vision.**

I. Introduction

The process of urbanization has resulted in a fast development of natural landscapes into constructed spaces, which has greatly diminished the availability of green spaces in the cities. The degradation of urban vegetation leads to a number of environmental problems, including air pollution, the urban heat island, the decrease of biodiversity, and the overall deterioration of the quality of life of the urban population. With the ever-growing cities, the issue of integrating green infrastructure in urban development, which is sustainable, is getting more and more attention by urban planners and environmental policymakers.

Parks, roadside plantations and urban forests are considered important in enhancing the environmental sustainability and the health of people. Nevertheless, it is not an easy task to find the appropriate locations where vegetation should be introduced in overpopulated cities. The classic methods of urban planning have traditionally been based on manual surveys, and stationary geographical data, which are sometimes time-consuming, subjective and inefficient when implemented on big urban surroundings.

New opportunities to automate environmental analysis and decision making have become available due to the recent advances in the field of artificial intelligence and computer vision. Specifically, Convolutional Neural Networks (CNNs) and other deep learning algorithms have shown miraculous results in the image analysis tasks like land-use classification, object localization and semantic segmentation. Such technologies can be used to interpret an aerial or ground-level image automatically, and it is possible to recognize patterns and spatial features that are hard to detect manually.

The current study hypothesizes the creation of an AI-based Urban Green Space Optimizer, the framework that will process the images of the urban territory and determine their capacity to be greened in a sustainable way. The suggested framework applies deep learning based image segmentation to point out the regions that can be planted in an input image and then computes the approximate area that can be used in the vegetation. The system also approximates the quantity of trees that can be planted in the identified region based on set rules of ecological spacing.

Besides giving visual information like plantable areas overlays, the system also produces amounts of information including plantable areas and projected planting capacity in quantitative form. The system will be accessible to planners, environmental researchers, and policymakers because it will allow them to upload images and retrieve instant results of the analysis through a web-based interface.

The main aim of the study is to show how artificial intelligence may be used to assist the sustainable urban development by offering intelligent decision-support systems

to green space planning. The proposed system will help to locate opportunities to increase urban greenery and foster environmentally friendly urban planning by employing the deep learning method, computer vision, and environmental modeling.

II. Literature Review

The study by Cañas et al. [1] presents an in-depth investigation of mapping of urban green spaces on a metropolitan level with very high-resolution satellite imagery and deep learning-based semantic segmentation. They used a U-Net network and utilized multiple CNN encoder backbones and tested the performance of segmentation on a large area of the city. Their model had the Intersection over Union (IoU) of 0.75 and the overall classification accuracy of 0.97, so the deep learning methods can be trusted when it is required to detect and outline green space boundaries in the dense urban setting. The paper has also shown that the pipeline was scalable to update the city-level green space inventories in accordance with SDG 11.7. Nevertheless, the scheme was fully reliant upon very high-resolution satellite data, and it lacked any projection of the variety of plants or trees that may be introduced to the designated areas.

The authors of the paper by Pešek et al. [2] provided a comparative analysis of a number of convolutional neural network designs to perform a semantical segmentation of urban green spaces based on freely available multispectral satellite images provided by Sentinel-2. The architectures they tested were FCN, U-Net [3], SegNet and DeepLabv3+ and the results were systematically analyzed in terms of the performance of the respective models in response to varying levels of urban land-use classes. Their results ensured that CNN-based segmentation was always more effective than the traditional image analysis methods when used on open-access multispectral data, and thus the method was viable and easy to repeat in urban green space studies. Although their segmentation methodology was quite robust, the work remained concentrated on the level of the satellite analysis, as well as it failed to introduce any site-specific planting estimation logic or interface that would allow the non-technical user to interact with the system.

Zhang et al. [4] investigated the association between the land surface temperature and the geographically weighted random forest and the configuration of the urban green space. Their report examined the effect on spatial features of green spaces, such as complexity of boundaries, size of individual patches, and their pattern of distribution, on the cooling effect on the immediate urban environment. The findings demonstrated that green spaces with more complex margins, broader single space and a clustered spatial distribution always generated a powerful mitigation of urban heat island effect as compared to broken or evenly dispersed green cover. Although this work was very informative on the effects of green space design on the temperature, it did not automate the process of identifying the appropriate areas to plant in a given land image, and produced no quantitative results on the planting capacity.

The article by Al-Rashidi et al. [5] used satellite images to evaluate the distribution of urban green space within the city of Riyadh with a machine learning-based framework. They used Sentinel-2 NDVI analysis and deep learning classification on their pipeline to approximate the level of tree density and green coverage at specific locations within urban areas. The analysis revealed that the machine learning based satellite based analysis as an alternative to the traditional field surveys could be used to assess the green spaces of cities on a city scale at a significantly faster and much cheaper rate. The work was, however, designed to analyze large scale regional analysis and could not be used to analyze individual plots at the site. It also lacked an interactive tool that planners or the local administrators could employ to evaluate the potential of the plantation of a particular area without the need to have remote sensing expertise.

Sharma et al. [6] conducted their reviews to encompass the deep learning methods available to land use and land cover (LULC) classification with the use of satellite imagery. It had a broad set of reviewed architectures, such as CNN, U-Net, LinkNet, and DeepLabV3+ and compared the performance of each to various types of data and classification tasks. The authors discovered that fully convolutional networks are especially effective in pixel-level segmentation in remote sensing, and that deep learning models have mostly substituted traditional feature extraction methods because of their high accuracy at complex and large scale data. In spite of having solid theoretical background to choose the segmentation architectures, the review was a mere survey contribution and made no suggestion on the implemented system to optimize green spaces or estimate the planting capacity. A recent review [7] further confirms that deep learning-based semantic segmentation has become the dominant paradigm for measuring urban greenness, yet most reviewed systems still operate at the satellite scale and lack end-to-end actionable planting estimation interfaces.

Patel et al. [8] proposed a green space design optimization framework, called DNN-GSOF, based on deep neural networks to be used in the planning of smart cities. The network was a changed U-Net structure that utilized a constraint-aware DC-GAN that took satellite pictures and sensor measurements of the environment and city map of the infrastructure. The system was meant to create design green space layout proposals that would not be in violation of the urban planning standards and optimizing ecological connectivity at the city level. Despite the fact that the outcomes were impressive in both design quality and space coherence, the architecture needed to have high-end GPU devices namely NVIDIA A100 accelerators, which are not applicable in small towns or low-resource settings. The interface was also not made easy to use by the users who lacked knowledge in deep learning.

Nguyen et al. [9] tested how deep learning could be used in semantic segmentation of aerial urban scenes in an 8-class land cover taxonomy which consisted of vegetation, roads, buildings, water bodies, and bare land. They contrasted DeepLabv3+ with a variation, AE-DeepLabV3+, and found

high segmentation accuracy on all categories, with the best performance of all the vegetation. Their multi-class segmentation scheme and preprocessing pipeline is directly applicable to the image processing part of the proposed system. Nevertheless, no downstream logic was implemented in the work to estimate the potential of planting and implement sustainability constraints and present the findings in a convenient user interface.

III. Research Gap

Based on the above reviewed studies, it is evident that the current research has come a long way in the detection and classification of urban vegetation using satellite and deep learning. Nevertheless, there is a uniform gap in all the reviewed literature: none of the works gives a full, end-to-end system, where a ground-level or aerial picture of a given land parcel is taken, plantable areas at the site level are delineated, and a quantitative and actionable planting recommendation is provided to a non-expert user. The majority of strategies are city or region-level and demand satellite imagery, some unique hardware, or remote sensing skills which are generally unavailable to local planners, municipal officers, or local communities. This gap is proposed to be bridged by the Urban Green Space Optimizer proposed, which integrates the CNN-based semantic segmentation with the spatial area estimation and a basic web-based interface that will enable intelligent plantation planning without the need of a satellite image, the system of the GPU infrastructure and knowledge of deep-learning algorithms.

IV. Proposed System and Methodology

A. System Overview

The Urban Green Space Optimizer that is proposed is an AI-based, image-based decision support system that is proposed to be used by urban planners, environmental officers, and community stakeholders to evaluate the planting potential of any open land parcel. The input to the system is one image of the ground level, a drone, or a low-altitude aerial image of an open field and the output is a series of quantitative markers of how much of the land can be plantable and the number of trees that can be successfully planted. This system was designed with the idea that it could be made available and at the same time be deep-technical. The system does not need satellite subscriptions, specialized GIS tools, or hardware with GPU capabilities, but instead needs to run on a regular machine and it is deployed via a lightweight web interface created using Streamlit. This has rendered it useful when it comes to use in municipal offices, schools and field surveys.

It is designed on a modular pipeline with every processing stage being independent and capable of being upgraded independently without any impact on other workflow stages. The pipeline takes input in raw image, preprocessing, CNN based segmentation, spatial area estimation and planting capacity calculation and finally gives results back to the user in the form of a visual overlay and numerical summary. This modularity also grants the capability to incorporate future environmental datasets like the quality of soil index and climate parameters that would be incorporated in the second phase of development.

B. Proposed Methodology

The proposed system methodology is designed in terms of a six-step pipeline, where each step will assist in a specific way in achieving the general objective of the estimation of sustainable planting capacity based on a land image. The steps are outlined below as per the order in which they are performed during a normal user session.

1) Image Acquisition and Upload: Image acquisition and upload commences when the user uploads a picture of the target land area using the web interface. The platform accommodates common image formats such as JPEG and PNG. The uploaded image is stored in an identified input folder in the project structure and the original file is stored as an audit and reanalysis in the future. At this point, the source of the images is not assumed, and the system can equally well work with both photos made on mobile gadgets, drones, or cameras mounted on the ground. Such flexibility has been an intended design strategy because most urban planners and municipal employees use photography equipment that is easily accessible instead of remote sensing tools.

2) Image Preprocessing: After the image is obtained, a preprocessing program is executed in order to prepare the image to be analyzed by CNN. The image is first converted back to RGB color space to conform to the normal deep learning input rules. It is then downsized to a constant resolution of 256 $\times$ 256 which is the size of the input that the segmentation model requires. Lastly, pixel values are brought to 0–255 range and converted to a floating-point range of 0–1 which stabilizes gradient calculation in inference and makes models consistently behave irrespective of the brightness or contrast of the original image. The image preprocessing utility module takes care of these steps and it is automatically processed without human interference.

3) CNN-Based Plantable Area Segmentation: The processed image is fed into a Convolutional Neural Network which is trained on binary semantic segmentation. The building is based on a lightweight autoencoder design, with the encoder consisting of consecutive convolutional and max-pooling operations to extract spatial information at different scales and the decoder consisting of upsampling layers which are used to rebuild the spatial resolution and generate pixel-wise prediction map. The last layer will add a sigmoid activation function that will produce a probability of 0–1 on each pixel indicating the potential that the similar spot on the original image will be a good site to plant. This probability map is then converted to a binary mask by thresholding with a value of 0.5 whereby any pixel with a value of 1 is considered as plantable and the rest of the pixels with 0 are considered as non-plantable. The decision to use a lightweight architecture instead of a heavier model like a full U-Net or DeepLabv3+ was to make sure that the system is able to be deployed on average hardware without the need to use any forms of GPU acceleration.

4) Plantable Mask Generation and Overlay: Once the binary segmentation mask has been generated it is scaled back to the original image size in order to enable the correct overlay rendering. The areas in the input image where it can be plantable are marked in green over a blended copy of the original (transparency weighted 70/30 original image/green overlay). This hybrid display enables the user to easily identify what areas on the land have been positioned as plantable, and still having sufficient detail of the original image to identify some of the roads, boundaries, structures, and other landmarks. The overlay image is stored to the processed outputs directory as well as viewed on the web interface that the user reviews.

5) Spatial Area Estimation: The area of plantable lands is measured by summing up all the pixels that are marked as plantable in the binary mask and translating the number into a real-life area value. The conversion factor of the pixel to a square meter is set to 0.25 square meters per pixel in the present implementation, which is a good approximation to the ground level images at a moderate altitude. This conversion factor can later be adjusted during the deployment where geo-referenced metadata or known information about a scale of the image is accessible. The calculated plantable area is returned to the user interface in the form of square meters as one of the key output measures.

6) Planting Capacity Estimation: This is the last step in the pipeline whereby the estimated area to be planted is now translated into a tangible planting suggestion. According to the general horticultural and urban forestry principles, it is expected that every tree should have a minimum distance of 9 square meters between them to ensure that roots have the space to grow, canopy to develop and air to move freely. The number of trees that can be planted sustainably is thus calculated as the integer division of the area that can be planted divided by the area that must be separated between the trees. This rule-estimation method was selected due to its transparency, interpretability, because users can make sense of it and, where necessary, modify the spacing assumption to suit other tree species or local planting rules. The output is displayed with the plantable area measurement and the visualization of the green overlay available to the user so that the user can obtain the complete view of the plantation potential of the land.

C. Design Decisions and Justification

A number of design choices in this system were made with measures to trade-off technical purity and practical utility. The simple CNN encoder-decoder as compared to a more advanced encoder-decoder such as Transformer-based segmentation models was based on the fact that the system should be runnable on commodity hardware. It was also due to the fact that the rule-based planting capacity estimator was chosen over the second ML model to provide interpretable output as planners must be able to interpret and have trust in the information before they can take action. The choice of developing the interface in Streamlit, as opposed to a full-stack web-based framework, was informed by the concept of minimum viable complexity, which permitted the system to be deployed and demonstrated within a very short time without the need to have a separate backend server and database stack. The decisions are a considered trade-off between state of the art performance and real world deployability, which is suitable to this stage of the project scope and purpose.

Training CNN segmentation model on the DeepGlobe [10] and LoveDA [11] datasets gave a total of 2279 labeled samples that were trained in 50 epochs with a maximum validation IoU of 0.7640 at Epoch 49. The segmentation engine loads the trained model weights at inference time which allows the system to give ecologically meaningful predictions of plantable areas on any given new input image. Section VII provides full quantitative evaluation results.

V. System Architecture

A. Architectural Overview

The Urban Green Space Optimizer system architecture is based on the three-tier layered system architecture, which consists of processing layer, presentation layer, and storage layer. The responsibility of each tier is well defined and each tier logs in to its neighbouring tiers using clearly defined interfaces. Such isolation of concerns also means that the modification of a single component of the system, like a replacement of the segmentation model or an improvement of the user interface, does not need to be reflected in the rest of the codebase. The general scheme here is deliberately maintained stateless at the session-level, that is, every upload of an image generates an independent cycle of processing independent of any stored data that might have been generated by earlier processing cycles. This architecture makes deployment easier and the system to be easily tested and debugged.

The topmost layer is the presentation layer, which is a local web application based on Streamlit, which is executed locally on the host machine and is hosted by a regular browser. The central processing layer, the middle layer, is the part that hosts three autonomously operating modules, i.e., the image preprocessing unit, the CNN-based segmentation engine, and the area and planting capacity estimation module. The lowest level is the storage layer, which supports all the file input and output activities keeping a distinct separation between raw input, processed intermediates and the ultimate output artefacts. Each of the layers and its internal components are described in detail as outlined below.

B. Presentation Layer — Streamlit Web Interface

The end user has direct contact with the presentation layer which is the sole component of the system. It is developed with the help of Streamlit, a Python open-source framework that translates Python scripts into an interactive web application without HTML, CSS, or JavaScript. The interface is in a single-page format and offers the user an image uploading button, an analysis triggering button and a tabular results panel. At this stage, when the user uploads a picture, Streamlit will write the file to the raw input directory and forward the file path to the processing layer. After the analysis occurs, the interface shows the results of three visuals simultaneously,

namely: the original image uploaded, the grayscale mask of the segmentation of the CNN, and the image with green overlays of the plantable area. Under the pictures, there are three metric cards with the calculated area planted in meters square, the number of trees to be planted estimated, and a sustainability density label. The whole interaction process, including image uploading up to displaying the result, is all in one application session and the user has no page navigation or file handling.

TABLE I
TECHNOLOGY STACK AND DEVELOPMENT ENVIRONMENT

Component	Tool / Library	Version	Purpose
Programming Language	Python	3.10	Core development
Deep Learning Framework	PyTorch	2.5.1	CNN model definition, training and inference
Image Processing	OpenCV	4.x	Image loading, color conversion, resizing
Numerical Computation	NumPy	Latest stable	Array operations, mask thresholding
Visualization	Matplotlib	Latest stable	Overlay rendering, mask saving
Web Interface	Streamlit	Latest stable	Browser-based user interface
Development Environment	VS Code	Latest	IDE with virtual environment support
Dependency Management	pip + venv	Python 3.10	Isolated package environment
Operating System	Windows 11	—	Development and testing machine
Hardware	NVIDIA GeForce RTX 3050 Laptop GPU	CUDA 12.1	GPU-accelerated model training and inference

C. Processing Layer — Core Modules

The processing layer has three modules invoked in that order whenever an analysis is being performed. The modules are used as independent Python files in the project directory and loaded into the main application script when running.

Image Preprocessing Module: This module is meant to convert the uploaded raw image in a format that the CNN model could understand. It reads the input file with the OpenCV, transforms the color space of BGR to that of RGB according to the common deep learning standards, resizes the image to 256 $\times$ 256 pixels to fit the input size of the model, and scales pixel values to the 0–1 range by dividing by 255. The module produces the original image of its natural size, which is overlay rendered, and the normalized image that is sent to the segmentation engine. By separately isolating these activities in a specific module the preprocessing logic can be tested and tuned separately without any alteration of the model or the interface.

CNN Segmentation Engine: This is a module in which the convolutional neural network is defined and instantiated in binary semantic segmentation. This is organized in the form of a symmetric encoder-decoder net. The encoder is composed of three convolutional blocks with one Conv2D layer and one MaxPooling2D layer that have a gradual downsizing of the spatial dimensions and an increment of the number of feature channels to 32 to 64 to 128 channels. This structure is replicated by the decoder by using two UpSampling2D layers and Conv2D layers which bring the channels down to 64 and then 32. The last output layer is a Conv2D layer with a single channel prediction map which is a single channel output layer with sigmoid activation and has the same spatial resolution as the model input. The Adam optimizer [12] with binary cross-entropy loss is used to compile the model, as it is suitable in terms of binary pixel-level classification. The segmentation engine is supposed to be stateless at inference time, by which it takes an image tensor as input and produces a prediction map without any internal state of session between calls.

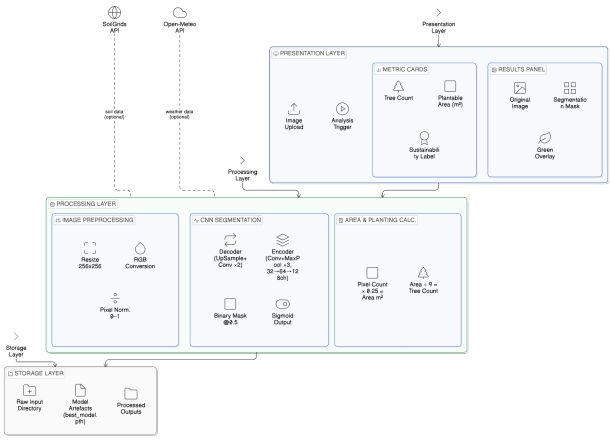

Fig. 1. System Architecture

Area and Planting Capacity Module (utils/area_and_plant_calculator.py): It is a module that will receive the saved overlay image as an input and will do two functions. The first one is to scan the image in grayscale followed by a binary threshold on pixel value 128 to isolate the mask of the plantable region and the total number of pixels classified as a plantable region. It then gives this number of pixels multiplied by the pixel-to-square-meter conversion of 0.25 to receive the number of square meters that can be planted. Second, it calculates the approximate number of trees dividing the amount of plantable land by the spacing that a tree needs which is 9 meters and giving the resultant integer number. The two outcomes are both relayed to the presentation layer. A parameter is used to accept the path of the overlay image in the module, which is why it is not difficult to replace the alternative source of the mask or change the conversion and spacing constants without redesigning the module.

D. Storage Layer — Project File Structure

The storage layer will be structured in a hierarchical directory structure with the raw inputs, model artefacts, utility scripts and processed outputs being grouped into distinctly labelled folders. This organization adheres to the standard

software engineering procedures of machine learning projects and makes sure that the project is navigable during the development stages of growth. The project root directory is named Urban_Green_Space_Optimizer and it has the following subdirectories and files.

VI. Results and Discussion

A. Experimental Setup

The model was tested on a special test set of 229 images which was taken on the combination of DeepGlobe and LoveDA datasets. The entire dataset was consisting of 2,279 samples (1,709 training, 341 validation, and 229 test) divided by a 75:15:10 ratio. The training was done on an NVIDIA GeForce RTX 3050 Laptop GPU having 4 GB VRAM and running CUDA 12.1 and PyTorch 2.5.1. The U-Net architecture was based on an encoder-decoder architecture and skip connections, as well as double convolutional blocks with Batch Normalization. Training was performed with 50 epochs with Adam optimizer in three phase learning rate: 1×10^{-4} during Epochs 1-32, 5×10^{-5} during Epochs 33-48 and 2.5×10^{-5} during Epochs 49-50. A Binary Cross-Entropy loss was combined with a Dice loss [13]. Inference was done on all the input images which had been resized to 256×256 pixels and normalized. The duration of total training was about 1 hour. Measures of evaluation Intersection over Union (IoU), Dice Score, Precision, and Recall measures were calculated at the pixel-level in comparison to binary ground truth masks at a threshold of 0.5.

B. Training Convergence

Table II contains a summary of training and validation at crucial epochs. The model has reached an improvement in validation IoU as Epoch 1 gave 0.6532 and the highest IoU was obtained at Epoch 49 (0.7640) which is stable and steady throughout the 50 epochs. The initial reduction of the learning rate at Epoch 33 resulted in a noticeable increase in validation IoU - the value at Epoch 30 was 0.7385 whereas it had reached 0.7557 at Epoch 33 - which validates the effectiveness of the scheduled decrease of the learning rate in finetuning the boundary-level segmentation in later epochs. Epoch 49 was the saved checkpoint which was used in all the test evaluations.

TABLE II
Training and Validation Metrics at Key Epochs. Best epoch (49) highlighted.

Ep.	Tr.Loss	Tr.IoU	V.Loss	V.IoU	LR
1	0.8549	0.5950	0.7057	0.6532	1×10^{-4}
5	0.6225	0.6872	0.6115	0.6672	1×10^{-4}
10	0.5662	0.7174	0.5602	0.7147	1×10^{-4}
20	0.5058	0.7417	0.5169	0.7363	1×10^{-4}
30	0.4725	0.7566	0.4944	0.7385	1×10^{-4}
33	0.4486	0.7701	0.4596	0.7557	5×10^{-5}
40	0.4300	0.7799	0.4586	0.7577	5×10^{-5}
49	**0.4098**	**0.7884**	**0.4358**	**0.7640**	$\mathbf{2.5 \times 10^{-5}}$
50	0.4070	0.7908	0.4321	0.7636	2.5×10^{-5}

C. Quantitative Evaluation Results

Quantitative evaluation results for the model on 229 image test sets are shown in Table III. A mean overlap of 0.6286 between the predicted plants in the masks and those that were manually identified by the research team and a Dice Score of 0.7254 indicate that there is moderate to strong agreement on the masks. An accuracy level of 0.7663 is indicative of good performance and accuracy for all pixels classified as plantable. An 0.7370 level of recall is indicative of successful recovery of most actual plantable areas in the test images.

TABLE III
Quantitative Evaluation Results on Test Set (229 images)

Metric	Score	Std Dev
IoU (Jaccard Index)	0.6286	$\pm$0.2978
Dice Score (F1)	0.7254	$\pm$0.2589
Precision	0.7663	—
Recall	0.7370	—
Best Validation IoU	0.7640	—

They have a relatively low standard deviation in the extent of IoU ($\pm$0.2978) which is a natural variation among the heterogeneous test set that covers dense urban scenes in LoveDA and open agricultural land in DeepGlobe. Images containing high imperviousness had lower IoU scores whilst the rural and semi-urban images with well-defined vegetation patches had IoU values of over 0.95 in multiple evaluation batches.

D. Qualitative Results

Fig. 2 indicates the raw satellite image that was uploaded to the application. The scene consists of a thick conglomeration of settlements in the middle with wide crop fields and vegetation belts.

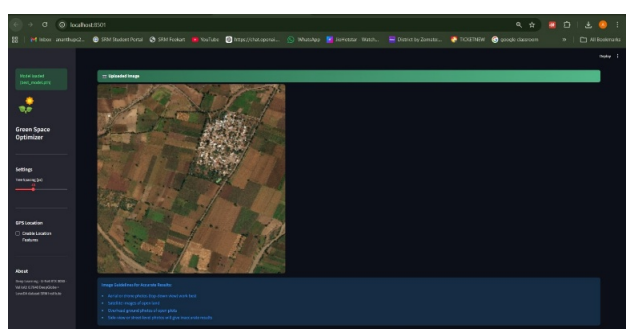

Fig. 2. Input satellite image uploaded to the Green Space Optimizer

Fig. 3 gives the view of before/after comparison. The right half has the image of the predicted plantable overlay of U-Net whereas the left half has the original image. Green areas represent the known areas of plantable quadriculated areas within the agricultural land whereas the red area appropriately denotes the thick built-up settlement area core as being non-suitable to be plantationed.

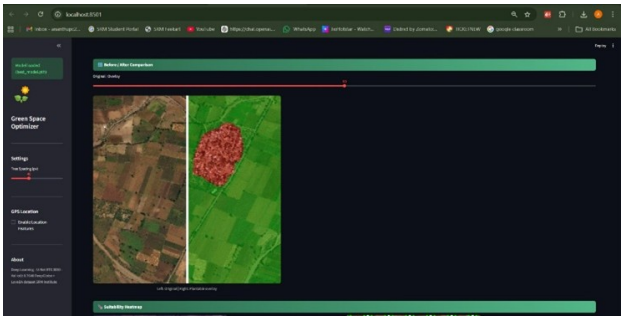

Fig. 3. Before/After overlay: Left — original; Right — plantable zones predicted by U-Net.

The heatmap of AI suitability and map of tree position are shown in Fig. 4. The heatmap validates the fact that the settlement core has the lowest score in the planting suitability whereas the open land has the highest score. With the tree placement map, it has been determined that there are 45 suggested locations of planting the trees. The Analysis Results panel shows that the area of the plantable land is 3,067 m^2, which supports 136 trees, 536 shrubs, and 1,533 small plants, it is estimated that the CO_2 will be sequestered at 7,069 kg/year and the Sustainability Index is 84/100.

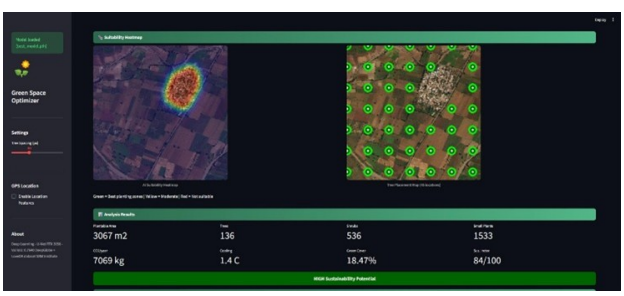

Fig. 4. AI suitability heatmap (left) and tree placement map with 45 locations (right), with analysis results panel.

Fig. 5 presents the full panel of results with the Greenery Assessment and Urban Intelligence results. The system rightly indicates that the area does not meet the 20% urban green standard (the current green cover is 18.47%), and they should be actively planted. Urban intelligence categorizes the region as Low Urban Density and High Plantation potential. The 9 m^2/person standard is achieved, but the standard of 20% green in urban planning is not achieved - a directly actionable result of the municipal planners. On the bottom of the panel, there is also a downloadable PDF report.

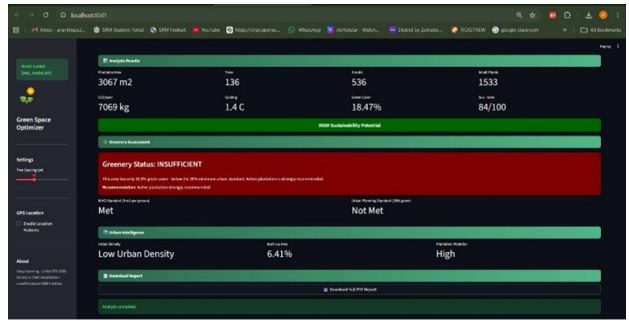

Fig. 5. Full results panel showing Greenery Assessment, Urban Intelligence classification, compliance checks, and PDF export.

E. Web Application Performance

The trained model checkpoint (best_model.pth) is loaded once when the application is started, and stored in memory to make inferences. On CPU-only deployment, approximately 3-5 seconds of end-to-end image upload through plantable mask generation, heatmap rendering, tree placement mapping and PDF report export. The application can take the standard JPEG and PNG aerial or satellite inputs. The main output parameters are: plantable area in square metres (PIXEL_TO_SQM = 0.05), recommended number of trees (16 m^2 per tree spacing), shrubs and small plants number, projected CO_2 capture in kg/year, approximate cooling in °C, green cover percentage, Sustainability Index (0-100) and WHO 9m^2 per person standard check, and urban planning 20 percent green cover standard check. Land-based applications When the user has GPS options enabled, GPS-linked soil (SoilGrids API) and weather (Open-Meteo API) context is provided to the location-aware planting advice.

F. Discussion

The test IoU of 0.6286 is reasonable considering it is trained in lightweight on binary vegetation segmentation without pre-training or transfer learning. An IoU of 0.75 on a similar binary segmentation task based on green space using U-Net on very high-resolution satellite imagery was reported in the study by Cañas et al. and per-class IoU values indicated by the U-Net-family models on Sentinel-2 multispectral data fall between 0.58 and 0.81 depending on the land cover type. The dissimilarity of validation and test IoU (0.7640 and 0.6286, respectively) indicates the presence of mild domain shift between the validation and test subsets, which may be attributed to the heterogeneity of the DeepGlobe and LoveDA source datasets in combination. Subsequent research will tackle this by dataset balancing, test-time augmentation and possible fine-tuning on India-specific urban satellite imagery of ISRO and Bhuvan open archives.

Combined, the quantitative and qualitative findings confirm the core system hypothesis: a U-Net which is trained on publicly available multi-source satellite images can provide high accuracy, useful plantable area estimates when used in urban green space planning, the output of which is good enough for preliminary municipal decision support. The outcome of deep learning segmentation, geospatial environmental

context, quantitative estimation of planting, and easy-to-use web interface is what makes this system remarkable compared to traditional NDVI-based or manual GIS systems that were reviewed in the literature.

VII. Conclusion and Future Work

A. Conclusion

This paper has provided an artificial intelligence-based decision support system, the Urban Green Space Optimizer, which recognizes plantable areas in urban land images by CNN-based semantic segmentation and predicts sustainable planting capacity by using spatial analysis. It was inspired by the fact that the urban vegetation in Indian cities is being rapidly degraded, and there are no available tools that enable the work of non-expert users to determine the potential of a plantation without subscribing to satellites, knowledge of GIS, or other sophisticated equipment. The system has been created in the format of six modules that include image acquisition, image processing, U-Net based segmentation, overlay visualization, spatial area estimation and planting capacity calculation. The U-Net model was trained using 2279 labeled samples in the DeepGlobe and LoveDA datasets through 50 epochs using an NVIDIA GeForce RTX 3050 Laptop GPU and it took about one hour to train. The three-phase learning rate schedule model with a best validation IoU of 0.7640 at Epoch 49 and a Dice Score of 0.7254 Precision of 0.7663 and Recall of 0.7370 on the 229-image held-out test set. These findings affirm that the model which is developed through training yields ecologically relevant predictions of plantable areas that are accurate enough to serve preliminary decision support at municipal levels. The web interface based on Streamlit provides end-to-end analysis of the plantable mask overlay, AI suitability heatmap, tree placement map, CO_2 sequestration estimate, the percentage of green cover, the Sustainability Index, the WHO standard compliance check, and a downloadable PDF report in a 3-5 second time on normal hardware without a need to have a graphics card. The system is directly aligned with three of the United Nations Sustainable Development Goals, namely SDG 11 (intelligent urban green space planning), SDG 13 (carbon sequestration estimation via vegetation planning), and SDG 15 (recovery and improvement of urban terrestrial ecosystems). Combined, the findings confirm that lightweight deep learning, as a tool to support spatial analysis, and a user-friendly interface can be used as a practical and scalable solution to environmentally responsible urban land management.

B. Future Work

Since the CNN segmentation model is now trained and validated, future research will be dedicated to scaling the system to the city scale by optimizing the green spaces by combining it with the Google Earth Engine to run large grids of overlapping satellite tiles over entire municipalities. This would facilitate city-wide potential maps of plantation rooms that emphasize neighbourhoods most abundant unused capacities of green space - a scale which would require a considerable amount of computational infrastructure and interaction with municipal GIS departments at scales bigger than the existing prototype.

A second long-term direction is retraining the segmentation backbone with a multi-class land cover framework that not only separates between plantable and non-plantable areas, but also the soil type, water closeness, and density of canopy as independent semantic classes that requires a significantly larger annotated dataset and multi-label training model.

The third extension that is proposed is that a temporal change detection module will be developed, which would compare previously taken satellite images of the same place over several years and calculate the loss of urban green cover over time that the system applied as the tool of urban environmental policy as well as a planning resource. Lastly, it will be integrated with IoT-based real time soil sensors to measure TDS, pH, and moisture on site to substitute the existing static SoilGrids API lookup with live ground-truth environmental validation, which will enhance the ecological accuracy of planting guidelines to be used in the field.

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